

Beyond purine restriction: dietary quality and sex differences in hyperuricemia

Joong Kyong Ahn

Division of Rheumatology, Department of Internal Medicine, Kangbuk Samsung Hospital, Sungkyunkwan University School of Medicine, Seoul, Korea

Hyperuricemia, characterized by elevated serum uric acid levels, is a biochemical precursor of various medical conditions, including gout, uric acid nephropathy, and the formation of uric acid stones. It is intricately associated with metabolic disorders (e.g., diabetes, obesity, and dyslipidemia), chronic kidney disease, and cardiovascular disease (e.g., hypertension, coronary artery disease, and stroke). Although the prevalence of hyperuricemia varies by region and ethnicity, the global burden has increased substantially over the past two decades [1,2]. Hyperuricemia and gout are more common in men than women; however, uric acid levels increase in women after menopause. Hyperuricemia should be understood as a condition affected by a complex array of factors, such as metabolic health, dietary patterns, renal efficiency, and hormonal dynamics, rather than being solely attributed to excessive food consumption.

In this issue of the *Journal of Rheumatic Diseases*, Lee and Kim [3] analyzed data from the 2019 to 2021 Korea National Health and Nutrition Examination Survey (KNHANES) involving a cohort of 14,305 individuals aged ≥ 20 years. Dietary quality was assessed using the Korean Healthy Eating Index (KHEI). This study employed the KHEI to thoroughly evaluate dietary quality, thereby providing a framework for discussing the relationship between overall dietary quality and hyperuricemia within Korean dietary practices. The KHEI is a comprehensive measure of dietary quality that includes 14 components, including breakfast, whole grains, fruits, vegetables, protein foods, dairy, saturated fat, sodium, sugars, and beverages, and the bal-

ance of carbohydrates, fats, and energy. In the real world, uric acid metabolism is influenced by complex dietary patterns, rather than by individual nutrients. This offers the advantage of evaluating hyperuricemia in the broader context of overall dietary patterns and metabolic health rather than solely attributing it to the intake of specific high-purine foods. Recent studies using the KHEI have consistently shown that overall dietary quality is associated with hyperuricemia. Using a large-scale KNHANES dataset, Son et al. [4] demonstrated that higher KHEI scores were associated with lower odds of hyperuricemia. Kim et al. [5] applied propensity score matching to further compare nutrient intake and dietary quality between individuals with and without hyperuricemia. Based on the KHEI evaluation, it was documented that individuals with hyperuricemia had significantly lower scores than the healthy controls. Nonetheless, the KHEI may not fully capture alcohol intake and specific subtypes of purine-rich foods. The reported studies collectively affirm the significance of the KHEI as a comprehensive dietary quality index in hyperuricemia research. Furthermore, the findings highlight the potential relevance of overall dietary quality in dietary counseling for individuals with or at risk of hyperuricemia.

Recent population-based studies have suggested a sex-dependent relationship between dietary quality and hyperuricemia [5-7]. Adherence to healthy dietary patterns, such as the Dietary Approaches to Stop Hypertension and Mediterranean diets, was associated with a reduced risk of hyperuricemia in both sexes. However, this protective effect appears to be more pronounced

Received August 21, 2026; Revised September 10, 2026; Accepted September 15, 2026; Published online September 17, 2026

Corresponding author: Joong Kyong Ahn,  <https://orcid.org/0000-0003-3246-4435>

Division of Rheumatology, Department of Internal Medicine, Kangbuk Samsung Hospital, Sungkyunkwan University School of Medicine, 29 Saemunan-ro, Jongno-gu, Seoul 03181, Korea. E-mail: mdahnjk@skku.edu

Copyright © The Korean College of Rheumatology.



This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (<http://creativecommons.org/licenses/by-nc/4.0>) which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited.

in men [7]. Kim et al. [5] reported that the link between sex and dietary quality is far more pronounced among men. Most available data originate from observational studies and systematic reviews, which indicate that the association between dietary patterns, overall dietary quality, and hyperuricemia demonstrates sex-specific differences [3,5,7-9]. Notwithstanding these earlier findings, a definitive consensus on sex-specific dietary recommendations remains elusive, largely because of the scarcity of randomized controlled trials that directly evaluate sex-based differences in urate-lowering efficacy. Consequently, well-designed, sex-specific clinical trials are crucial for providing the evidence required for sex-specific dietary recommendations.

Lee and Kim [3] underscore the importance of sex-specific analysis as a primary research question. Their work makes a significant contribution by employing separate stratified analyses, rather than merely treating sex as a covariate, to reevaluate the relationship between dietary quality and hyperuricemia in Korean adults. In men, higher KHEI scores were associated with a gradual reduction in the risk of hyperuricemia, whereas no significant trend was observed in women. In a multivariable logistic regression model with restricted cubic splines, men showed a consistent reduction in risk as KHEI scores increased, whereas women exhibited a U-shaped risk pattern [3]. This study demonstrates that improving the overall dietary quality may serve as a potentially modifiable factor associated with lower odds of hyperuricemia in men. In contrast, the findings highlighted the complexity of uric acid metabolism in women, which cannot be solely attributed to dietary factors. Although these findings do not warrant distinct dietary recommendations for men and women, they strongly support the need for sex-stratified analyses and personalized interpretations in future dietary research on hyperuricemia. Although Lee and Kim's study [3] demonstrates considerable strengths, its findings should be interpreted cautiously owing to its inherent limitations, including the cross-sectional study design, recall bias, and insufficient adjustment for confounding factors. Nonetheless, utilizing a large-scale Korean cohort, this study holds significant clinical value by reaffirming the potential relevance of overall dietary quality to hyperuricemia while highlighting the necessity of a sex-specific approach. These findings suggest the importance of considering sex when interpreting the relationship between dietary quality and hyperuricemia. In men, dietary patterns were directly correlated with the risk of hyperuricemia. In contrast, in women, non-dietary factors, such as hormonal shifts, meta-

bolic status, medication, and renal function, are likely to exert a greater influence. Therefore, these findings shift the discussion from generic dietary considerations toward a more nuanced, sex-specific interpretation of the association between dietary quality and hyperuricemia. Ultimately, these findings should be viewed as hypothesis-generating evidence supporting sex-stratified dietary research. Future prospective cohort studies and randomized controlled trials are required to determine whether personalized, sex-specific dietary interventions can improve long-term clinical outcomes.

FUNDING

None.

ACKNOWLEDGMENTS

None.

CONFLICT OF INTEREST

J.K.A. has been an editorial board member since May 2024 but has no role in the decision to publish this article.

REFERENCES

1. Ngandeu-Singwe M, Nkeck JR, Hamroun A, Endomba FT, Ngoufack-Tientcheu C, Ndoadougue AL, et al. Worldwide trends in hyperuricaemia from 2000 to 2023: a systematic review and modelling analysis. *Lancet Rheumatol* 2026;8:e346-62.
2. Ahn JK. Epidemiology and treatment-related concerns of gout and hyperuricemia in Korean. *J Rheum Dis* 2023;30:88-98.
3. Lee JW, Kim Y. Sex-specific differences in associations between diet quality and hyperuricemia in Korean adults. *J Rheum Dis* 2026;33:205-15.
4. Son H, Lee HS, Lee SY, Lee W. Association between the Korean Healthy Eating Index and hyperuricemia among Korean adults. *J Korean Med Sci* 2026;41:e135.
5. Kim EY, Heo SJ, Lee Y, Han TH, Kwon YJ, Lee JW. Nutrient intake and diet quality in individuals with hyperuricemia: a matched population study. *Front Nutr* 2026;13:1719104.
6. Wang H, Qin S, Li F, Zhang H, Zeng L. A cross-sectional study on the association between dietary inflammatory index and hyperuricemia based on NHANES 2005-2018. *Front Nutr* 2023;10:1218166.
7. Juraschek SP, Gelber AC, Choi HK, Appel LJ, Miller ER 3rd. Effects of the Dietary Approaches to Stop Hypertension (DASH) diet and sodium intake on serum uric acid. *Arthritis Rheumatol* 2016;68:3002-9.

8. Wen ZY, Wei YF, Sun YH, Ji WP. Dietary pattern and risk of hyperuricemia: an updated systematic review and meta-analysis of observational studies. *Front Nutr* 2024;11:1218912.
9. Cheng S, Shan L, You Z, Xia Y, Zhao Y, Zhang H, et al. Dietary patterns, uric acid levels, and hyperuricemia: a systematic review and meta-analysis. *Food Funct* 2023;14:7853-68.